

AVINASH UMMAGANI

Software Engineer | Android

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TECHNICAL SKILLS

Languages: Kotlin, Java, Python, JavaScript (working knowledge)
Android & AOSP: Android SDK, AOSP System Apps, Jetpack Compose, Soong Build System
Architecture: MVVM, Modular Architecture, Monolithic-to-Modular Migration
Build & Tooling: Gradle, Soong, Repo, Android Studio, VS Code, Figma
Media & Real-Time Comms: ExoPlayer, Glide, WebRTC, OpenTok, Zoom SDK
Networking & DI: Retrofit, OkHttp, Volley, Gson, Hilt, Coroutines
Data & Persistence: Room, SQLite, ViewModel, LiveData
Notifications & Monitoring: Socket.io, MQTT, Firebase Crashlytics
Testing & Debugging: Appium, Robot Framework, LeakCanary
Version Control: Git, GitHub, Bitbucket, Gerrit

PROFESSIONAL EXPERIENCE

Software Engineer, Caavo Labs India Pvt Ltd – Bengaluru, India (Remote) **Apr 2022 – Present**
Remote Caregiving System

- **Modular architecture migration:** Migrated a legacy monolithic Android codebase to a multi-module architecture, with modules built using Soong and Gradle
- **IPC & shared state:** Built a content provider-based state service for cross-app data sharing, with hands-on experience across Android services, AIDL, and broadcast receivers for inter-process communication
- **System app & device control:** Developed system-level Android apps for device onboarding on an Android 11 (AOSP) device, using the AOSP TV Input Framework, and built universal remote control of connected source devices via IR, IP, CEC, and BT
- **Real-time communication:** Integrated OpenTok and Zoom SDKs for video and audio calling, and used WebRTC for remote TV control. Customized and built the WebRTC Android library from source, adding a custom audio device module to share custom audio data
- **Smart plug integration:** Added BLE and MQTT-based smart plug integrations, along with HDMI-CEC, to monitor TV power state
- **Data persistence:** Designed a Room/SQLite persistence layer for local data storage
- **Media pipeline:** Built ExoPlayer-based media playback and Glide-based image loading
- **Cross-platform exposure:** Contributed a couple of features to a React Native codebase
- **Notifications, analytics & stability:** Delivered a TV-friendly push notification service and an in-house analytics pipeline with network-aware batched uploads, plus on-device bug reporting and Firebase Crashlytics for diagnostics
- **Platform collaboration:** Partnered with platform and hardware teams through Gerrit code reviews on AOSP system app changes

Systems Engineer, Tata Consultancy Services (TCS) – Hyderabad, India **Jul 2021 – Apr 2022**
Enterprise CRM application

- **Feature development:** Delivered new features and resolved bugs on a large-scale enterprise CRM application
- **Agile delivery:** Coordinated with QA and backend teams in Agile sprints to ship consistent, stable releases

EDUCATION

B.Tech, Electronics & Communication Engineering — VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad | CGPA: 9.5 / 10 **2021**